Power, phasors and complex power

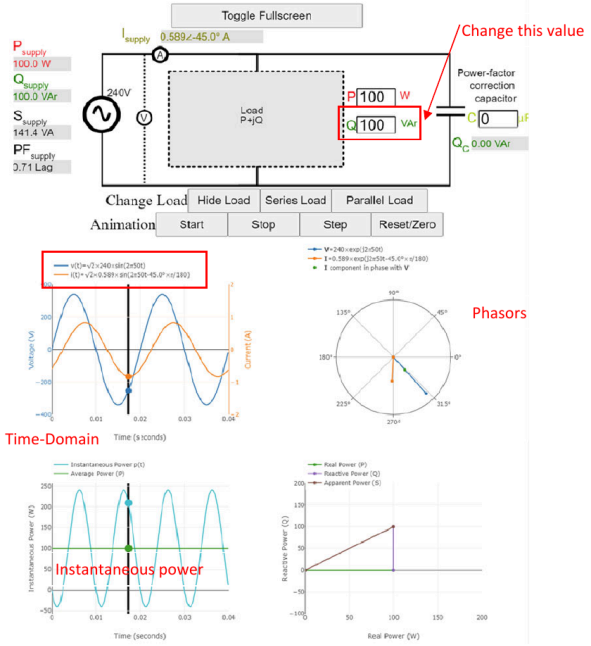
Practice questions · Day 01 · Sat 24 Oct · pack M1

5. Practice

From your course files

Power Circuit Tool WS1 Q1 (workshop). *Time-Domain Waveforms:* Using the default values of load P = 100 W and Q = 100 VAr, examine the equations for $v(t)$ and $i(t)$ which are given just above the voltage and current time waveforms.

a. What are the peak and rms values of the voltage and current waveforms?
b. For each of the values of Q = 100 VAr, 0 VAr and −100 VAr: (i) indicate whether the current is leading, lagging or in-phase; (ii) indicate whether the power factor is leading, lagging or unity.



Power Circuit Tool WS1 Figure 1: the tool interface, with the default load P = 100 W, Q = 100 VAr on 240 V (from the workshop sheet)

Power Circuit Tool WS1 Q2 (workshop). *Phasors:* examine the relationship between the phasor graph and the time-domain graphs.

a. Which direction are the phasors rotating?

b. Does the x-axis or y-axis component of the phasors correspond to the instantaneous voltage and currents? You may find it convenient to use the animation control buttons: start, stop, step and reset/zero.
c. Reset the phasors to stop them rotating. As you change Q = 100 VAr, 0 VAr and −100 VAr, how do you work out whether the current is leading, lagging or in-phase with the voltage?

Power Circuit Tool WS1 Q3 (workshop). *Instantaneous Power:* the instantaneous power waveform $p(t) = v(t) \times i(t)$ is shown in blue in the graph below the time-domain waveforms.

a. What is the frequency of the instantaneous power? Hint: find the period over which it repeats (T) and hence find its frequency ($f = 1/T$). How does this frequency compare to that of the voltages and currents?
b. Under what circumstances is the instantaneous power zero?
c. For the values of Q = 100 VAr, 0 VAr and −100 VAr: (i) What is the change in the average instantaneous power (shown in green)? (ii) What is the (approx.) minimum value of the instantaneous power? Note that negative values of instantaneous power are associated with reactive power. We will cover this in the next workshop.

PQ 1.14 (practice questions). Calculate the following quantities

a. Torque when the mechanical power is 7.5 kW and the speed is 2910 rpm.
b. Mechanical power when the speed is 50,000 rpm and the torque is 4.25 Nm
c. Speed (in rpm) when the power is 30 MW and the torque 191 kNm

PQ 1.21 (practice questions). Consider a single-phase AC machine. The input voltage is 110 V, the input current is 15 A and the power-factor is 0.8 lagging. The output speed is 1750 rpm and the output torque is 6 Nm. Calculate the:

a. input power
b. output power
c. efficiency
d. power loss

PQ 1.22 (practice questions). Consider a three-phase AC machine. The voltage of each phase is 240 V and the power-factor is 0.85 lagging. The output speed is 650 rpm and the output torque is 63 Nm. It is known that the machine efficiency is 88%. Calculate the:

a. output power
b. input power
c. power loss
d. input current magnitude

PQ 2.6 (practice questions). A certain car battery can be modelled using an equivalent circuit of a 12.5 V voltage source with a series resistance of 0.02 Ω. Calculate:

a. The open-circuit battery output voltage.
b. The battery current and internal power loss under short-circuit conditions.
c. The battery output voltage and current when powering a load of 1 ohm.
d. When the starter motor is energised, the battery output voltage is observed to fall to 8 V. Calculate the current drawn by the starter motor and the power supplied to the starter motor.
e. Calculate the maximum output power available from the battery.

PQ 2.15 (practice questions, do (f)–(h)). Calculate the magnitude and phase of the current in the load connected to a single-phase 240 V AC source. You can assume the phase angle of the voltage is zero. The load consists of:

a. a 10 ohm resistor
b. a 24 ohm resistor
c. a 24 ohm resistor in series with a 10 ohm resistor
d. a 24 ohm resistor in parallel with a 10 ohm resistor
e. an inductor with a 24 ohm reactance
f. a 24 ohm resistor in series with an inductor with a 24 ohm reactance
g. a 24 ohm resistor in parallel with an inductor with a 24 ohm reactance
h. the load in part (a) in parallel with the load in part (f). [Hint: you can add the currents from (a) and (f)]

PQ 2.17 (practice questions). A single-phase load consists of a parallel-connected 25 ohm resistor and an inductor with a 125 ohm reactance is connected to the 240 V mains, calculate

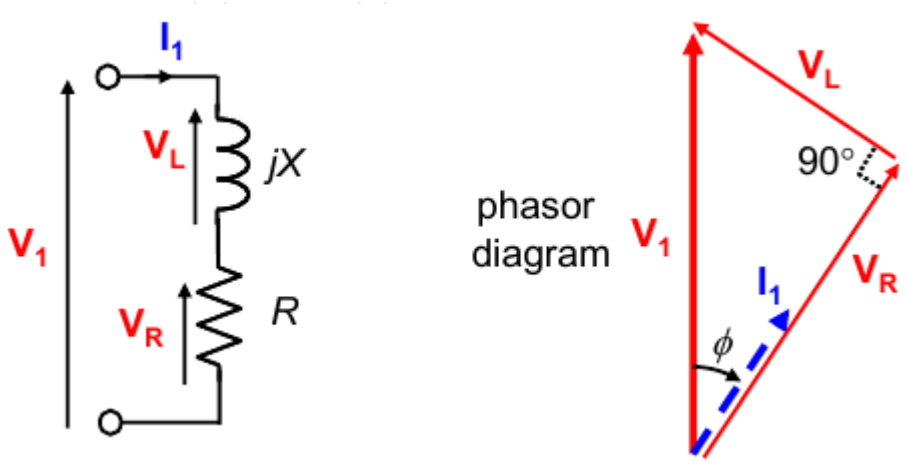
a. the load impedance Z in polar form
b. write the supply voltage $\mathbf{V_1}$ in phasor form (assume it has a phase angle of zero)
c. the resultant phase current phasor $\mathbf{I_1}$
d. the input power factor (remember to show lead or lag)
e. the complex power S_1 in both polar and Cartesian form
f. the input real power P using (i) the real part of S_1; (ii) the input power factor and V_1 and I_1; (iii) the fact that all the power losses occur in the resistor and hence the losses are V^2/R (why not I^2R?)
g. the input reactive power Q
h. the input VA (that is, the apparent power S)
i. sketch a power triangle and verify that $P^2 + Q^2 = S^2$
j. sketch a phasor diagram showing the relationship between the voltage and the current

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PQ 2.18 (practice questions). In the following circuit, it is known that V_1 (magnitude of $\mathbf{V_1}$) = 200 V and V_R = 150 V.

a. Find V_L.
b. Find the power-factor angle and power-factor of the circuit.
c. If the input current magnitude I_1 is 10 A, find the input apparent, real and reactive power. Use this input current magnitude for the following parts of this question.
d. What is the real and reactive power absorbed by the resistor?

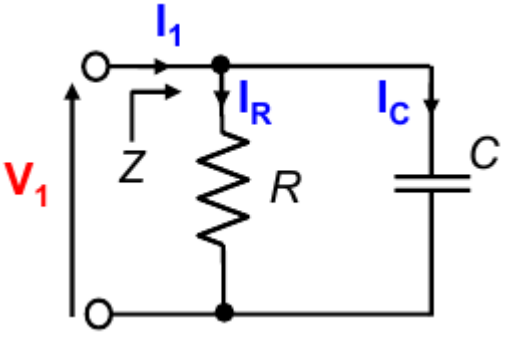
e. What is the real and reactive power absorbed by the inductor?
f. Using the results from (d) and (e), what is the resistance and reactance of the circuit?



PQ 2.18 figure: series R and jX with V_R, V_L and V_1, and the phasor diagram (from the practice questions)

PQ 2.19 (practice questions). In the following circuit, is known that the input voltage V_1 is 140 V, the input power P_1 is 500 W and the input current I_1 is 15 A.

a. Find the value of the resistance R.
b. Find the input apparent and reactive power.
c. Find the power-factor angle ϕ and hence the power-factor.
d. Show that you get the same result for the power-factor using pf = P/S.
e. Using the value of reactive power, find the value of reactance.
f. Sketch a phasor diagram for the circuit showing $\mathbf{V_1}, \mathbf{I_1}, \mathbf{I_R}, \mathbf{I_C}$ and ϕ. It is not necessary to show any values for the above phasors in the diagram.
g. Sketch a power triangle for the circuit showing the values of S, P and Q and ϕ.



PQ 2.19 figure: parallel R and C on V_1, with branch currents I_R and I_C (from the practice questions)

Jun23 Q1a (past exam, 5 marks). *Battery Modelling.* The equivalent circuit on the right is used to model a car battery.

i. The battery's open-circuit voltage is measured at 13.8 V. Find the value of the internal voltage source E. [1 mark]
ii. A DC starter motor is connected to the battery. It draws a current of 300 A and the car battery voltage falls to 9.5 V. Find the value of the internal resistance of the battery, R_S, the power dissipated in this resistance, and the power delivered to the motor. [3 marks]
iii. Find the battery output voltage when a load drawing a current of 20 A is connected to it. [1 mark]

Jun23 Q1a figure: battery model E in series with R_S, terminal voltage V, load R_L (from the exam paper)

Jun23 Q1c (past exam, 9 marks). *Phasor Calculations.* Consider the single-phase series RLC load as shown in the figure on the right. The current $\mathbf{I_1}$ is known to be $13\angle+20^\circ$ A.

i. Using the known current $\mathbf{I_1}$, find the voltage phasors across the resistance $\mathbf{V_R}$, inductance $\mathbf{V_L}$ and capacitance $\mathbf{V_C}$ in polar form using Ohm's law. [3 marks]
ii. Find the phase impedance Z of the load in polar form and hence the voltage phasor $\mathbf{V_1}$ in polar form using Ohm's law. [3 marks]
iii. The complex power S is given by $S = \mathbf{V_1}\mathbf{I_1}^*$, where * refers to the complex conjugate. Find the complex conjugate of the current $\mathbf{I_1}^*$, thus the complex power S and hence the total real power P and reactive power Q supplied from the power source. [3 marks]

Jun23 Q1c figure: series j30 Ω, 10 Ω and −j50 Ω carrying I_1, with V_L, V_R, V_C and V_1 marked (from the exam paper)

Jun24 Q1a(ii) (past exam, 2 marks). *Fill-In-The Blank.* An electric motor has an input power of 1000 W and a power loss of 300 W. This results in an output power of _____ W and an efficiency of _____ %. Increasing the power loss for the same input power would cause the efficiency to _____ (choose decrease or increase). Increasing the input power for the same power loss would cause the efficiency to _____ (choose decrease or increase).

Jun24 Q1b (past exam, 3 marks). *Electrical and Mechanical Power.* Consider a 3ph, 4 pole, 50 Hz induction motor. Under a certain operating condition, the motor speed is 1420 rpm, input voltage is 415 V, input current is 6.8 A, input electrical power is 4 kW and output torque is 23 Nm.

i. Find its input power factor (you can assume this is lagging). [1 mark]
ii. Find its output power and hence efficiency. [2 marks]

Jun24 Q1c (past exam, 5 marks). *AC Circuit Analysis.* Consider the series resistor-inductor (RL) circuit shown on the right.

i. The input current I_1 has a magnitude of 2 A. Find the value of the real power and reactive power absorbed by the resistor. Find also the real and reactive power absorbed by the inductor. [3 marks]
ii. For a different value of input current I_1, it is known that the resistor voltage V_R has a magnitude of 50 V and the inductor voltage V_L has a magnitude of 20 V. Hence find the magnitude of the input voltage V_1. [2 marks]

Jun24 Q1c figure: series j40 Ω and 100 Ω carrying I_1, with V_L, V_R and V_1 marked (from the exam paper)

New questions

N1. Single-phase pump motor [5 marks]. A single-phase motor runs from 230 V, 50 Hz and draws 6.2 A at a pf of 0.78 lagging. It delivers 0.95 kW to its load at 2,860 rpm.

a. Find the input real power and the efficiency. [2]
b. Find the output torque. [1]
c. Find the input apparent and reactive power. [2]

Circuit for New question 2: 30 Ω in series with 100 µF on 240∠0° V, 50 Hz

N2. Series RC [6 marks]. A 30 Ω resistor in series with a 100 µF capacitor is connected to 240 V, 50 Hz. Take $V_1 = 240\angle 0^\circ$ V.

a. Find the impedance Z in polar form. [2]
b. Find the current I_1 and the voltages V_R and V_C as phasors. Show that $V_R^2 + V_C^2 = V_1^2$. [2]
c. Find S, P, Q and the power factor. [2]

Circuit for New question 3: the 230 V, 5.0 A, 900 W load as series (a) and parallel (b) RL models

N3. One measurement, two models [6 marks]. An inductive load on 230 V, 50 Hz draws 5.0 A and 900 W.

a. Find the apparent power, reactive power and power factor. [2]
b. Find R, X and L if the load is modelled as series RL. [2]
c. Find R, X and L if the load is modelled as parallel RL. [2]

Circuit for New question 4: motor load A in parallel with load B (40 Ω in series with −j30 Ω) on 240 V, 50 Hz

N4. Two loads on one supply (chained) [8 marks]. A 240 V, 50 Hz single-phase supply feeds two loads in parallel. Load A is a motor delivering 1.5 kW of mechanical output, with an efficiency of 80 % and a pf of 0.75 lag. Load B is a 40 Ω resistor in series with a capacitor of reactance 30 Ω.

a. Find the electrical P, S and Q of load A, and the magnitude of its current. [3]
b. Find the current phasor of load B and its complex power. [2]
c. Find the total P, Q and S, the supply pf and the supply current magnitude. [3]

Circuit for New question 5: series R = 8 Ω, L = 40 mH, C = 200 µF on 230∠0° V, 50 Hz

N5. Series RLC with a surprise [7 marks]. A series circuit of $R = 8\ \Omega$, $L = 40$ mH and $C = 200$ µF is connected to $V_1 = 230\angle 0^\circ$ V, 50 Hz.

a. Find X_L, X_C and Z (polar). Is the circuit inductive or capacitive overall? [2]
b. Find I_1, V_R, V_L and V_C. Comment on $|V_C|$. [2]
c. Find S, P, Q and the pf using $S = V_1 I_1^*$, and check P and Q from the element values. [2]
d. At what supply frequency would the pf be unity? [1]

N6. Three-phase motor nameplate (brief preview of M2) [6 marks]. A 3ph induction motor is rated at 415 V, 7.5 kW, 1,455 rpm, with an efficiency of 88 % and a pf of 0.84 lag. Its connection is not stated.

a. Find the rated torque. [1]
b. Find the rated input power and losses. [2]
c. Find the rated line current and the total input reactive power. [2]
d. Find the phase voltage. [1]

6. Quick recall check

1. A UK supply is 230 V rms. What is its peak value?
2. Convert 3000 rpm to rad/s, and 100 rad/s to rpm.
3. A load has $P = 1200$ W and $Q = -500$ VAr. Is its pf leading or lagging, and what is the value?
4. Series RL: $V_R = 80$ V and $V_L = 60$ V. What is V_1?
5. Why can't you add the apparent powers of two parallel loads?